Morteza Akbari

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About

Artificial Intelligence researcher, with experience in computer vision, graph neural networks, and medical imaging. My master's research led to the FARFUM-RoP dataset, published in *Scientific Data*. I have since worked on breast ultrasound segmentation, multi-camera tracking, and retrieval-augmented vision-language models. More recently, I have been developing spatio-temporal models for satellite-based poverty estimation through a research collaboration, combining temporal attention with spatial graph learning. I am interested in developing methods that learn from visual, spatial, and temporal information, with a particular focus on computer vision and multimodal learning.

Education

Ferdowsi University of Mashhad, M.Sc. in Computer Science, Artificial Intelligence — Sept 2021 – Sept 2023

- GPA: 17.17/20.0 (Ranked 4th among 17 students, Top 25%)
- **Thesis:** Deep learning-based ROP disease classification with the interpretation of decision-making
- **Supervisor:** Dr. Hamid Reza Pourreza

University of Birjand, B.Sc. in Computer Science — Sept 2017 – Sept 2021

- GPA: 18.86/20.0 (Ranked 1st among 120 students, Top 1%)
- **Thesis:** Handwriting recognition from noisy images using image processing methods
- **Supervisor:** Dr. Hamid Saadatfar

Publications

Hierarchical Attention with Graph-Aware Transitions for Robust Multi-Camera Multi-Target Tracking
Akbari M., et al. Journal of Information Systems and Telecommunication (JIST), 2026 (Under Review)

Edge-Enhanced Graph Attention for Relation-Adaptive Spatio-Temporal Multi-Camera Tracking
Akbari M., et al. Journal of Information Systems and Telecommunication (JIST), 2026 (Under Review)

A combined UNet++ and LSTM approach for breast ultrasound image segmentation
Hesaraki S., Akbari M., et al. Franklin Open, 2025. DOI: 10.1016/j.fraope.2025.100385

FARFUM-RoP, A dataset for computer-aided detection of Retinopathy of Prematurity
Akbari M., et al. Scientific Data (Nature Portfolio), 2024. DOI: 10.1038/s41597-024-03897-7

Study and evaluation of the application of artificial intelligence and neural networks in the diagnosis of retinopathy of prematurity
Akbari M., et al. The First International Conference on Information Technology, Management and Computers, 2024

Research Experience

Multi-Camera, Multi-People Tracking, University of Birjand — 2023 – Present

- Designed edge-enhanced Graph Attention Network for cross-view identity association, achieving 10% reduction in ID errors and substantially reducing identity switches
- Implemented and benchmarked framework in PyTorch with CUDA acceleration
- Submitted 2 journal papers (under review) from project results

Spatio-Temporal Graph Neural Networks for Poverty Estimation from Satellite Imagery, Research Collaboration — Feb 2026 – Sept 2026

- Develop spatio-temporal models for poverty estimation using Landsat and nighttime-light satellite embeddings, with attention-based LSTM modeling of multi-year temporal patterns.
- Construct hybrid spatial graphs based on geographic proximity and satellite-history similarity, incorporating gated graph correction and rural–urban edge filtering.
- Train and evaluate models using African Demographic and Health Surveys (DHS) data with geographically disjoint splits, conducting ablation studies to assess temporal and graph components.

Retrieval-Augmented VLM for Document Image Understanding, University of Birjand — 2025 – 2026

- Contributing to RAG system integrating visual and textual modalities for document image understanding
- Exploring cross-modal alignment between image and text domains for visual question answering
- Optimizing prompt-engineering strategies and constructing knowledge graphs to improve interpretability

Hybrid UNet++–LSTM Model for Breast Tumor Segmentation, University of Birjand 2023 – 2025

- Conducted formal analysis and data curation for a hybrid UNet++–LSTM framework for breast ultrasound image segmentation
- Analyzed the integration of multi-scale feature extraction, LSTM-based temporal modeling, and attention mechanisms for breast tumor segmentation
- Evaluated the proposed model on the BUSI dataset, achieving 98.88% accuracy, 91.20% sensitivity, and 92.74% Dice coefficient

Deep Learning for Retinopathy of Prematurity Classification, Ferdowsi University of Mashhad 2021 – 2023

- Led design and implementation of complete deep learning pipeline for ROP detection, achieving >92% test accuracy with EfficientNet models
- Created and published FARFUM-RoP dataset (Nature Scientific Data 2024), establishing new benchmark
- Implemented model interpretability techniques (Grad-CAM, LIME, SHAP) improving clinical trust

Work Experience

Part AI Research Center, Computer Vision Engineer Feb 2022 – May 2023

- Conducted applied research focusing on multi-object tracking, tiny face detection, and low-resolution face enhancement within single-camera systems
- Designed deep learning–based tracking pipelines, motion modeling, and re-identification modules, achieving a 12% increase in tracking accuracy and reduced ID-switch rates
- Researched tiny face detection algorithms and also face super-resolution and enhancement models to restore facial details from low-resolution inputs
- **Tools:** Python, TensorFlow, YOLO, DeepSORT, ByteTrack, BoT-SORT, RetinaFace, TinyFace, GFPGAN, ESRGAN, FSRNet

Technical Skills

Programming Languages: Python, MATLAB, C++, C, Java

Machine Learning and Deep Learning: Transformers, GNNs, Attention Mechanisms, YOLO, DeepSORT, ByteTrack, BoT-SORT, RetinaFace, TinyFace, GFPGAN, ESRGAN, FSRNet

Frameworks, Libraries and Tools: PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, Git, CUDA, GeoPandas, Matplotlib, LangChain, HuggingFace Transformers, NetworkX, Linux, LaTeX

Awards and Honors

- **Top-Ranked Graduate Student**, M.Sc. in Computer Science (Artificial Intelligence), Ferdowsi University of Mashhad - Rank 4 among 17 Students (2021–2023)
- **Full Tuition Scholarship** for M.Sc. studies, Ferdowsi University of Mashhad (2021–2023)
- **Ranked top 1% nationally** in the Computer Science graduate entrance exam - Rank 96 among 10,000+ applicants (2021)
- **Top-Ranked Graduate Student**, Bachelor in Computer Science, University of Birjand - Rank 1 among 120 Students (2017–2021)
- **Full Tuition Scholarship** for Bachelor studies, University of Birjand (2017–2021)

References

Dr. Adel Daoud
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